Yang Yang

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Education

University of Southern California *Aug 2025 – Present*
PhD in Mechanical Engineering | GPA: 3.94/4.00

Sichuan University *Sep 2021 – Jun 2025*
B.Eng. in Mechanics with Honors | GPA: 3.81/4.00

Selected Publications and Manuscripts

†Equal contribution. Full list: Google Scholar.

- S. Li†, T. Wu†, J. Xu†, Y. Huang, Z. Zhang, H. Zhao, Q. Xu, Z. Wang, L. Ye, **Y. Yang**, C. Lyu, S. Mu, X. Wang, Z. Xie, C. Wu, X. Yu, W. Ding. "Biomimetic multimodal tactile sensing enables human-like robotic perception." *Nature Sensors*, vol. 1, pp. 52–62, 2026. Webpage | PDF | Video
- **Y. Yang**, T. Zhang, Y. Wang, W. Yue, T. Liu, H. Ren. "Conformable Vision-Based Tactile Sensor with Enhanced Soft Elastomer Design for Palpating Irregular Anatomical Surfaces." *International Conference on Biomimetic Intelligence and Robotics 2025 (ICBIR)*. Webpage | PDF | Video
- T. Zhang†, **Y. Yang**†, Y. Yang, H. Gao, J. Lai, H. Ren. "Three-dimension Tip Force Perception and Axial Contact Location Identification for Flexible Endoscopes using Tissue-compliant Soft Distal Attachment Cap Sensors." *ICRA*, pp. 538–544, 2025. Webpage | PDF | Video
- S. Li†, J. Xu†, T. Wu, **Y. Yang**, Y. Chen, X. Wang, W. Ding, X.-P. Zhang. "Vitire: A Bimodel Visuotactile Tire with High-Resolution Sensing Capability." *IEEE/ASME Transactions on Mechatronics*, 2025. Webpage | PDF | Video
- **Y. Yang**, B. Zhang, H. Wu, Y. Zhang, H. Zhang, Y. Liu, Q. Wang. "A deep learning approach for low-cycle fatigue life prediction under thermal–mechanical loading based on a novel neural network model." *Engineering Fracture Mechanics*, vol. 306, Art. no. 110239, 2024. Webpage | PDF
- X. Huang†, **Y. Yang**†, S. Tang, Z. Chen, Y. Jin, M. Li, L. Yang, J. Wu, D. Song, H. Zhao. "Morphable Nerve Cuffs for Neural Stimulation and Recording on Sub-mm Peripheral Nerves." *Manuscript in preparation.* Video
- C. Zhang, S. Yin, X. Huang, **Y. Yang**, X. Zhang, Q. Zhao, W. Gao, H. Zhao. "Strong, switchable electroadhesive clutches from microphase-separated ionoelastomers for multifunctional robotic interfaces." *Nature Communications*, in revision. Video

Technical Skills

Programming: Python, C++, PyTorch, MATLAB

Robotics & Sensing: Robot Learning Algorithms, Manipulation, Tactile Representation, Pressure Sensor Array Algorithms, Isaac Lab, ROS, Gazebo, Linux/Ubuntu

Design & Simulation: SolidWorks, AutoCAD, Finite Element Analysis, Sensor Calibration

Hardware: STM32 Microcontrollers, Arduino, Robot Arm, ATI Force Sensors, 3D Printing, Laser Cutting

Microfabrication: Sputtering, Reactive Ion Etching, Plasma Cleaning

Research Experience

CoTac: Co-located Electro-Visual Tactile Sensor *Feb 2026 – Present*
University of Southern California | Doctoral Research (Ongoing) | Advisor: Hangbo Zhao

- Designing a hybrid tactile sensor combining resistive pressure sensing and visual deformation cues
- Developed dual-modality markers for local normal-pressure sensing and camera-based motion tracking
- Developing 3D force decoupling and distributed force-field reconstruction from the co-located measurements

Wireless Wall-Climbing Robot with Electroadhesive Clutches *Sep 2025 – Dec 2025*
University of Southern California | Robot Design and Integration | Advisor: Hangbo Zhao

- Designed and 3D-printed a quadruped crawler to demonstrate electroadhesive wall locomotion
- Integrated collaborator-developed clutches at the feet and implemented STM32-based robot control
- Enabled wireless operation through Bluetooth communication between the robot and a computer

Morphable Nerve Cuffs for Neural Stimulation and Recording *Mar 2026 – Present*
University of Southern California | Co-first Author | Advisor: Hangbo Zhao

- Co-developed a pressure-controlled neural cuff with 0.5–1.6 mm diameter tuning to accommodate nerve size variation and simplify deployment
- Fabricated complete cuff devices, from PDMS preparation and laser-cut pouch fabrication to PEDOT electrode design, buckle assembly, and encapsulation
- Conducted in vivo animal experiments to evaluate neural stimulation and recording with the morphable cuffs

Tactile Perception in Surgical Procedures *Oct 2024 – May 2025*
The Chinese University of Hong Kong | Research Assistant | Advisor: Hongliang Ren

- Developed a conformable vision-based tactile sensor for palpation of irregular anatomical surfaces
- Built sensing, imaging, and illumination modules, calibrated the sensor, and implemented 3D reconstruction
- Simulated FBG endoscope sensor loading to evaluate radial/axial force sensing and axial contact localization, contributing to a co-first-author ICRA 2025 paper

Simulation and Learning for Tactile Sensing *Jun 2024 – Aug 2024*
Tsinghua University | Summer Research Intern | Advisor: Wenbo Ding

- Generated simulated tactile datasets with marker force and displacement labels to support contact-force estimation
- Used a neural network to predict force distributions from binary elastomer images and developed real-time marker tracking and 3D reconstruction
- Simulated visuotactile tire loading to relate deformation to force, contributing to the SuperTac and Vitire studies

In-situ Parameter Identification for GelSlim *Jun 2023 – Aug 2023*
Shanghai Jiao Tong University | Summer Research Intern | Advisor: Daolin Ma

- Investigated in-situ mechanical parameter identification for GelSlim to characterize tactile sensing elastomers
- Generated an FEA dataset and used a Transformer to infer mechanical parameters from boundary conditions
- Validated the approach in real-world experiments, with errors of 0.03 MPa for Young's modulus and 0.006 for Poisson's ratio

Projects

Tactile Dexterous Manipulation via Reinforcement Learning *Sep 2026 – Present*
Robot Learning Course Project | University of Southern California

- Developing reinforcement learning methods for tactile manipulation with a dexterous robotic hand

LeRobot-based Robotic Arm *Jul 2026 – Present*
Independent Project

- 3D-printed and assembled a LeRobot-based arm as a platform for manipulation experiments

Lightweight Earthquake-Resistant Structure *Sep 2023 – Mar 2024*
Undergraduate Project | Sichuan University

- Designed, modeled, and simulated a lightweight earthquake-resistant structure, then fabricated a physical prototype from bamboo strips

Teaching Experience

Teaching Assistant, Dynamics *Fall 2026*
University of Southern California

Teaching Assistant, Structural Mechanics *Spring 2024*
Sichuan University

Academic Service and Honors

Reviewer, ICRA, AJRIS

USC Viterbi School of Engineering Graduate School Fellowship *2025*

Top 100 Undergraduate Students of Sichuan University *2025*

First Prize, Sichuan Province Mechanics Competition (Individual; Group Team Leader) *2023*